Little Explorers · Class 22

Button Melody Maker

Press a button, play a note, make a song!

High & low | Button notes | Play a tune



Did you know?
Electronic keyboards are just lots of buttons, each one telling a computer to play a note!

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Let's wonder...

Can we build our own piano?

Music from buttons

A piano has lots of keys. Press one: a note! Could we make a tiny piano with just 3 buttons, a buzzer and a computer?



do re mi

Too hard!

Need a big piano?

Let's build it!

Today: you build a **3-button melody maker** that plays do, re, mi.

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Learning objectives

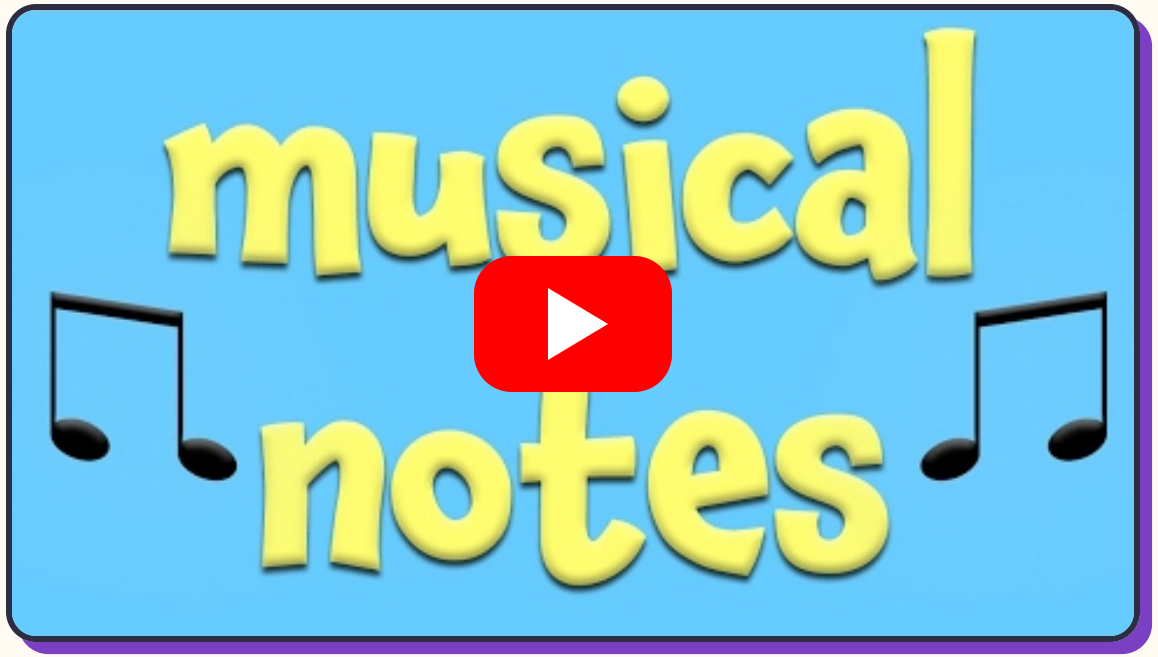
Today I can...

1	**High and low:** hear and say if a note is **high** or **low**	**Proof:** I sorted 3 notes into high and low
2	**Buttons play notes:** make each **button** play a different note	**Proof:** my 3 buttons play 3 notes
3	**Play a tune:** play a **short tune** with my buttons	**Proof:** I played the start of a song

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Objective 1 · High and low · video

Video time!



Musical Notes! Learning about music

KLT Wild · about 4 minutes · youtu.be/YjFIlLKjmkI

Watch for...

1. What are the musical notes called?
2. Which notes sound **high** and which sound **low**?
3. How do notes make a song?

Scan to watch

After the video: Sing do-re-mi going UP like stairs. Then come back DOWN!

Objective 1 · High and low

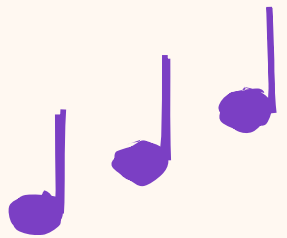
High notes and low notes

The idea

Some sounds are **high**, like a bird singing. Some are **low**, like a big drum or a lion's roar.

Notes go up like stairs: **do, re, mi, fa, so, la, ti, do!**

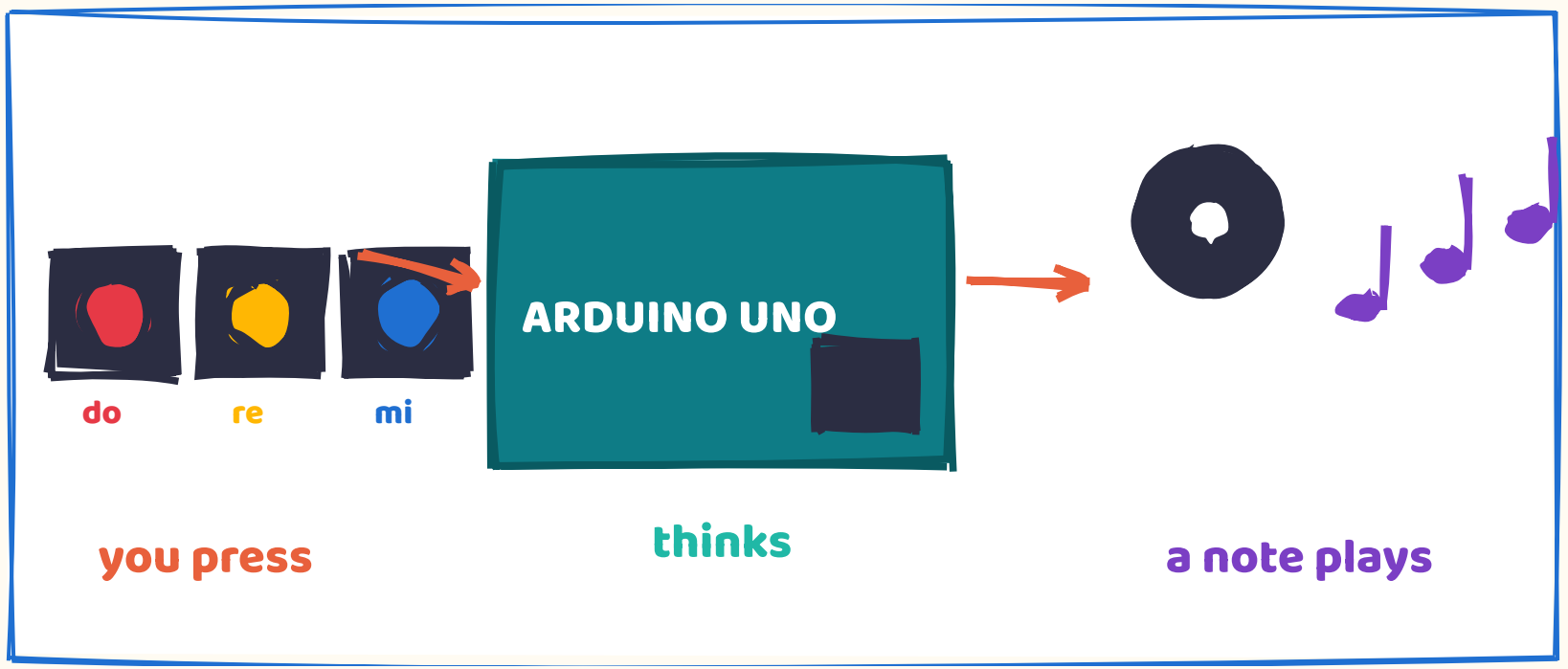
Example: a mouse and an elephant



A mouse squeaks with a high voice. An elephant rumbles with a low voice. Can you do both voices?

Our machine

Press, think, play



How it works

You **press** a button.
The Arduino **thinks**: which button? Which note?
The **buzzer** sings that note!

Teacher: parts are pre-wired on the breadboard. Children plug in the USB cable, snap the picture blocks in mBlock and test.

Objective 2 · Buttons play notes

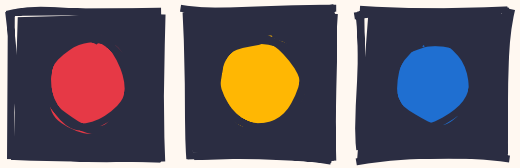
Each button = one note

The idea

The computer checks every button many times a second. When one is pressed, it plays **that** button's note.

Press red, yellow, blue: **do, re, mi!** Stairs going up!

Example: doorbells



do re mi

Different doors can have different doorbells. Each bell rings its own sound. Our buttons each ring their own note.

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Objective 3 · Play a tune

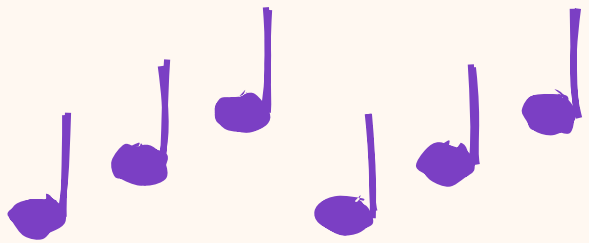
A tune is notes in order

The idea

Songs are notes in a special **order**, with some notes **repeated**.

'Mary had a little lamb' starts: **mi re do re mi mi mi**. Can you play it with your buttons?

Example: talking



Words in a sentence must be in order to make sense. Notes in a song must be in order to make a tune.

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Real world

Buttons that make music



Keyboards
Every key is a button



Phone ringtones
Computers play note patterns



Toys
Musical toys with buttons



Doorbells
Ding dong: two notes!



What other things in your home play a little tune?

Fun fact
An ice-cream van's song is a list of notes played by a tiny computer!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. High or low?
2. Snap the picture blocks
3. Play a tune!
4. Music challenges

Class game

High or low?

How to play

1. Teacher plays a note on a phone, xylophone or the buzzer.
2. High note: stretch up tall like a giraffe!
3. Low note: crouch down like a frog!
4. Then two notes: which one is higher?

Then discuss

Was it easier to hear high notes or low notes?

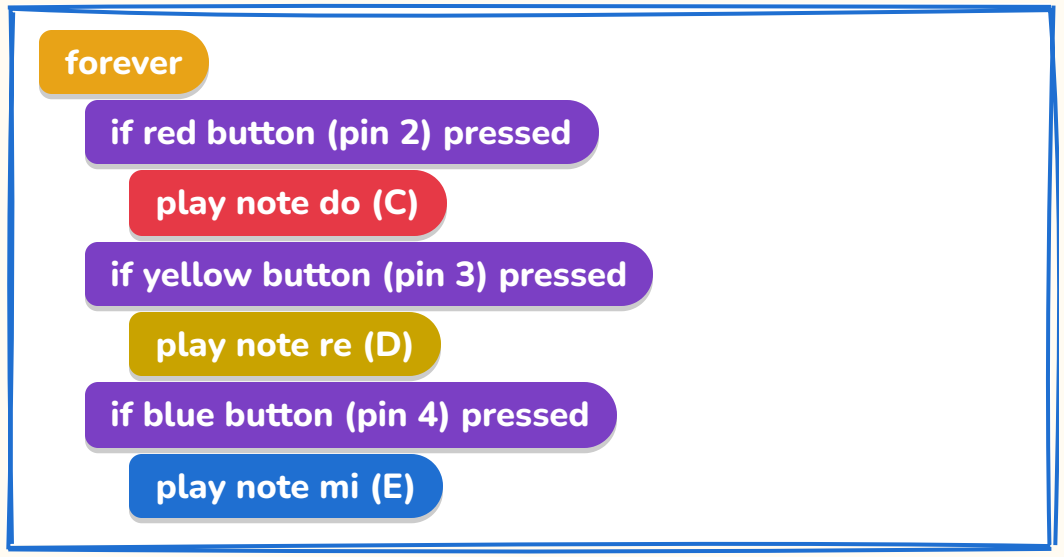


Big idea

Our ears can tell high from low. Computers can make any note we ask for.

Time to build!

Snap the picture blocks



Read it like a story

If red is pressed: do.

If yellow: re.

If blue: mi.

Check again and again!



Teacher

Buttons are pre-wired to pins 2, 3, 4 with the buzzer on pin 8. Use mBlock or ArduBlock picture blocks.

MAKE

Music time!

Play a tune!

Song	Buttons	Did it work?
Mary had a little lamb	mi re do re mi mi mi	_____
Going up	do re mi	_____
Going down	mi re do	_____
My own tune	_ _ _ _ _	_____

Talk about it

Which tune was easiest? Make up your own and teach a friend!

Colour code

Red = do, yellow = re, blue = mi.

MAKE

Level up!

Music challenges

Choose your challenge

- **Fourth button:** add 'fa' (F) on a new button.
- **Long notes:** make a note last longer.
- **Song button:** one button plays a whole tune!
- **Band:** play with another team at the same time.

Think like an engineer

Why does the computer need to check the buttons again and again?

Boss level

Play 'Twinkle Twinkle' with 6 notes!

Objective check

Can I really do it? Prove it.

01	Is this note high or low? Show me!	○ ○ ○
02	Press each button. Which note does each play?	○ ○ ○
03	Play the start of a tune for us.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Class concert: each team plays one tune!

Show what you know

What did we learn?

- [x] Notes can be high or low
- [x] Each button plays its own note
- [x] If a button is pressed, play the note
- [x] A tune is notes in order

At home

Sing do re mi at home, going up and down like stairs!



Exit ticket: which is higher: a bird or a lion?

Next class:
Colour Detective Sensor!